

Supplement: Where Query-Conditioned Audio Ranking Transfers—and Where It Fails

Author metadata required before submission

I. CONTROLLED AND EXTERNAL EXACT-ONSET RESULTS

The source-isolated panel contains 34,193 chunks from 2,956 unique source files and has zero development source overlap. The external panel excludes every UrbanSound8K Freesound ID found in ESC-50 metadata and uses LibriSpeech test-other backgrounds. Its relative AP and Hit@1 intervals exclude zero, but the prespecified absolute performance index is 72.77 rather than 85.

II. FROZEN NATURAL OUTCOME

The complete-prediction sensitivity to pinned upstream ActionFormer sorting and tIoU arithmetic has maximum mAP difference 0.00001873. Reproducing both upstream semantics locally matches within 5.55e-17 and leaves every metric-dependent gate unchanged.

III. CAP-CORRECTED CROSS-FITTING DETAILS

Each candidate pool retains at most 50 segments per class before shared video capping. Reapplying the frozen sort and 200-segment video cap reconstructs all ten original candidate files exactly. For held fold f and class c , the candidate is selected on all folds except f ; class streams are joined only after selection and capped once.

The full-versus-fixed-CLAP paired video interval is [5.8165, 6.6397] percentage points. The post-hoc incremental full-versus-CLAP-only interval is [1.1571, 1.5895] points. These video-macro effects are not algebraically identical to global class-macro mAP differences.

IV. ALL NATURAL CLASS EFFECTS

Every full-router class mean AP is above fixed multiscale CLAP. Every full-router class is also above the CLAP-only router, although several increments are very small. This is descriptive post-outcome evidence, not ten independent confirmatory tests.

V. DOWNSTREAM ANSWER BOUNDARY

The learned-minus-prefix difference is -1.9444 percentage points with 95% source-bootstrap interval [-5.2778, 1.4111]. AUDITA remains sealed and is not used to repair the failed bridge.

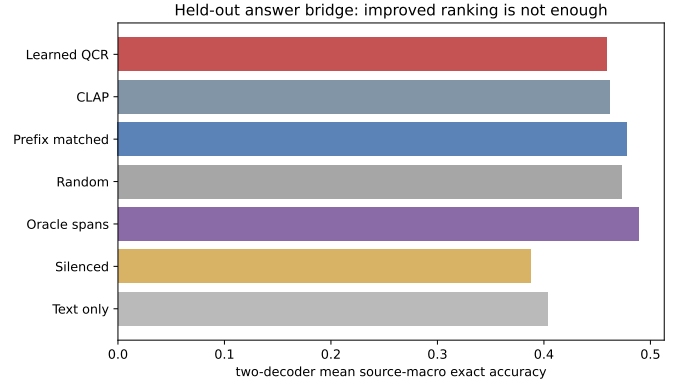


Fig. 1. Two-decoder mean source-macro exact answer accuracy. Oracle spans are context, not an eligible learned comparator.

VI. INVALIDATIONS AND AMENDMENTS

- The first cross-fit proposal joined per-class outputs that had each already passed a video cap. This could exceed 200 segments after joining. The analyzer was stopped before a router report; the observed candidate mAP values and defect are retained in `INVALIDATION_OUTPUT_CAP.md`.
- The corrected track rebuilds uncapped per-class pools, reconstructs all ten source caps exactly, routes classes, and applies one shared cap.
- Before router scoring, a quadratic per-video cap assertion was replaced by an equivalent single-pass counter. The amendment changes no candidate, selection, metric, bootstrap, or cap parameter.
- Candidate-pool files are large regenerable intermediates and remain outside the compact submission ZIP. Their exact row counts and SHA-256 values are included. Final full and CLAP-only predictions are included.

VII. REPRODUCTION MAP

- `q1_plus/`: controlled confirmation and answer-bridge reports, frozen checkpoints, and relevant analysis code.
- `q1_90/`: external exact-onset protocol, recipes, report, and acquisition/rights records.
- `q1_top_tier/`: natural freeze, prediction reports, metric audit, supervised contextual receipt, and source.
- `q1_crossfit_capcorrect/`: cap-correct locks, candidate receipt, final predictions, report, tests, and publication analysis.
- `figure_source_data/`: source rows for every figure.

TABLE I
FROZEN AND DIAGNOSTIC CONJUNCTIONS.

Stage	Gate	Result
Controlled exact-onset	AP gain ≥ 0.05 ; Hit@1 gain ≥ 0.05 ; both bootstrap lower bounds > 0 ; at least four positive seeds; no condition Hit@1 loss below -0.05 ; integrity	7/7 pass
External exact-onset	Controlled gates plus no class Hit@1 loss below -0.05 and performance index ≥ 85	7/8; index 72.77
Frozen natural transfer	global delta > 0 ; video lower bound > 0 ; at least four positive thresholds; worst class ≥ -0.05 ; QCR multi $\geq$ CLAP 4 s; integrity	2/6 fail
Cap-correct diagnosis	global delta > 0 ; video lower bound > 0 ; threshold breadth; class non-harm; 10 exact cap reconstructions; 50 assignments; integrity	7/7 met
Answer bridge	gain over strongest and random; backbone non-harm; response/content and integrity checks	5/8 fail

TABLE II
EVIDENCE-RANKING METRICS. BOOTSTRAP INTERVALS ARE PAIRED AT THE RECIPE OR ORIGINAL-SOURCE UNIT DECLARED BY EACH PROTOCOL.

Panel	Recipes	Prior AP	QCR AP	AP difference (95% interval)	Prior Hit@1	QCR Hit@1
Source-isolated confirmation	397	0.6090	0.8425	0.2335 [0.1850,0.2841]	0.7683	0.9446
External source-disjoint	219	0.5408	0.7020	0.1612 [0.0911,0.2334]	0.6256	0.7534

TABLE III
NATURAL ZERO-SHOT GATE OBSERVATIONS.

Gate	Observed	Decision
Global mAP difference > 0	-1.4973 pp	Fail
Video-bootstrap lower > 0	-1.4749 pp	Fail
Positive thresholds ≥ 4	0/5	Fail
Worst class ≥ -5 pp	-7.1301 pp	Fail
QCR multi $\geq$ CLAP 4 s	0.052603 vs. 0.052280	Pass
Integrity	zero failures	Pass

TABLE IV
FOLD-BY-CLASS SELECTIONS. C DENOTES CLAP, Q DENOTES QCR, AND MS DENOTES POOLED MULTISCALE.

Class	Candidate family	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4
Human:Speech	CLAP+QCR	Q-4	Q-4	Q-4	Q-4	Q-4
Human:Speech	CLAP-only	C-MS	C-MS	C-MS	C-MS	C-MS
Animal:sound	CLAP+QCR	Q-4	Q-4	Q-4	Q-4	Q-4
Animal:sound	CLAP-only	C-4	C-4	C-4	C-4	C-4
Object:opening/closing of something	CLAP+QCR	C-2	C-2	C-2	C-2	C-2
Object:opening/closing of something	CLAP-only	C-2	C-2	C-2	C-2	C-2
Object:sound	CLAP+QCR	C-2	C-2	C-2	C-2	C-2
Object:sound	CLAP-only	C-2	C-2	C-2	C-2	C-2
Human:Sound(clapping)	CLAP+QCR	C-0.5	C-0.5	C-0.5	C-0.5	C-0.5
Human:Sound(clapping)	CLAP-only	C-0.5	C-0.5	C-0.5	C-0.5	C-0.5
Interaction: fluid	CLAP+QCR	C-2	C-2	C-2	C-2	C-2
Interaction: fluid	CLAP-only	C-2	C-2	C-2	C-2	C-2
Interaction:dropping	CLAP+QCR	Q-1	Q-1	Q-1	Q-1	Q-1
Interaction:dropping	CLAP-only	C-1	C-1	C-1	C-1	C-1
Interaction:hitting	CLAP+QCR	C-1	C-1	C-1	C-1	C-1
Interaction:hitting	CLAP-only	C-1	C-1	C-1	C-1	C-1
Interaction:rolling	CLAP+QCR	C-1	C-1	C-1	C-1	C-2
Interaction:rolling	CLAP-only	C-1	C-1	C-1	C-1	C-2
Object:breaking of something	CLAP+QCR	C-4	C-4	C-4	C-4	C-4
Object:breaking of something	CLAP-only	C-4	C-4	C-4	C-4	C-4

TABLE V
OUTPUT-CAP AUDIT.

Route	Pre-cap rows	Final rows	Discarded	Videos over 200	Max pre-cap
CLAP+QCR	1,453,945	1,044,156	409,789	4,401	356
CLAP-only	1,662,296	1,069,264	593,032	5,119	389

TABLE VI
CLASS MEAN AP AND CROSS-FIT DECOMPOSITION.

Class	Fixed CLAP	CLAP-only	Full	Full-fixed	Full-CLAP-only
Human:Speech	0.082197	0.082996	0.101187	0.018990	0.018191
Animal:sound	0.005356	0.021934	0.026341	0.020985	0.004407
Object:opening/closing of something	0.010656	0.015561	0.016328	0.005673	0.000767
Object:sound	0.041209	0.057493	0.059273	0.018064	0.001780
Human:Sound(clapping)	0.199148	0.279395	0.279408	0.080260	0.000014
Interaction: fluid	0.134348	0.248483	0.248969	0.114621	0.000487
Interaction:dropping	0.022112	0.034267	0.034968	0.012855	0.000701
Interaction:hitting	0.132861	0.147388	0.151216	0.018355	0.003828
Interaction:rolling	0.045023	0.045280	0.045487	0.000464	0.000207
Object:breaking of something	0.002854	0.006413	0.006772	0.003918	0.000359

TABLE VII
HELD-OUT ANSWER RESULTS.

Retriever	Exact accuracy
Learned QCR	0.4586
CLAP	0.4619
Prefix matched	0.4780
Deterministic random	0.4731
Oracle spans	0.4886
Silenced selected spans	0.3877
Text only	0.4033